

Melika Nobakhtian

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RESEARCH INTERESTS

My research centres on **natural language processing** and **large language models**, with a particular focus on their **evaluation**: cultural knowledge, reasoning, and behaviour in low-resource languages such as Persian. I am also interested in **trustworthy** language models — interpretability, alignment, and safety — as well as controllable text generation, creativity in AI, and multimodal and agentic NLP systems.

EDUCATION

M.Sc. in Computer Engineering — Data Science

Sep 2023 – Feb 2026

Khatam University, Tehran Institute for Advanced Studies (TEIAS), Tehran, Iran

- GPA: 4.0 / 4.0 (19.25 / 20)
- Best Data Science Student Award, 2025

B.Sc. in Computer Engineering — Artificial Intelligence

Sep 2018 – Jul 2023

Iran University of Science and Technology (IUST), Tehran, Iran

- GPA: 3.87 / 4.0 (18.04 / 20)
- IUST ranked 4th among Iranian universities (QS 2023)

Diploma in Mathematics and Physics

2014 – 2018

Farzanegan High School (NODET — National Organization for Development of Exceptional Talents)

- GPA: 4.0 / 4.0

PUBLICATIONS

1. **Beyond Meter: Benchmarking Formal Compliance, Lexical Diversity, and Aesthetic Quality in Persian Poetry Generation.** *Under review, ACL ARR August 2026 cycle, 2026*
2. **CommonLID: Re-evaluating State-of-the-Art Language Identification Performance on Web Data.** *ACL 2026 (Main Conference), 2026*
3. **Evaluating Cultural Knowledge and Reasoning in LLMs Through Persian Allusions.** *Findings of EMNLP 2025, 2025*
4. **NICA at EXIST CLEF Tasks 2024.** *CLEF 2024, 2024*
5. **A Silent Voice: A Bilingual Longitudinal Analysis of Mahsa Amini's Movement Tweets.** *ICDMW 2024, 2024*
6. **IUST_NLPLAB at ImageCLEFmedical Caption Tasks 2023.** *CLEF 2023, 2023*
7. **IUST at ImageArg: The First Shared Task in Multimodal Argument Mining.** *ArgMining @ EMNLP 2023, 2023*
8. **IUST_NLPLAB at ImageCLEFmedical Caption Tasks 2022.** *CLEF 2022, 2022*
9. **ParsVQA-Caps: A Benchmark for Visual Question Answering and Image Captioning in Persian.** *WiNLP @ ACL 2022, 2022*

RESEARCH EXPERIENCE

Research Assistant — Large Language Models Laboratory

Apr 2026 – Present

Tehran Institute for Advanced Studies (TEIAS), Khatam University | Advisors: Dr. Mohammad Taher Pilehvar & Dr. Yadollah Yaghoobzadeh

- Research on the **evaluation and benchmarking of LLMs**, probing cultural knowledge, commonsense reasoning, and performance in low-resource settings such as Persian.
- Working on **Persian poetry generation and evaluation** with LLMs, benchmarking formal compliance, lexical diversity, and aesthetic quality — submitted to ACL ARR, August 2026 cycle.

- Contributing to peer-reviewed publications and reviewing for NLP/ML venues.

Research Assistant — Natural Language Processing Laboratory

Jun 2021 – Jun 2024

Iran University of Science and Technology (IUST) | Advisor: Dr. Sauleh Eetemadi

- Conducted research in **multimodal and low-resource NLP**: medical image captioning and visual question answering, multimodal argument mining, and Persian benchmark construction.
- Built and curated datasets (crawling, cleaning, annotation) and implemented deep-learning pipelines that led to first-place shared-task results and multiple publications.
- Applied NLP to **computational social science**, including bilingual longitudinal analysis of social-media discourse.

Research Assistant — Advanced Big Data Analysis Laboratory

Jun 2021 – Sep 2021

Iran University of Science and Technology (IUST) | Advisor: Dr. Hossein Rahmani

- Studied foundational data-science methods and libraries; contributed to backend development for data-analysis tooling.

HONORS & AWARDS

- **Best Data Science Student Award (2025)** — recognised as the outstanding Data Science student at TEIAS, Khatam University.
- **Persian LLM AI Hackathon (2024)** — finalist in the first round; **4th place** in the final stage with the IUST-LLM team.
- **ImageCLEF — Cross-Language Image Retrieval Track (2022)** — **1st place** in the ImageCLEFmedical Caption Prediction subtask with the IUST_NLPLAB team.
- **M.Sc. Admission without Entrance Exam, IUST (2022)** — admitted directly for ranking among the top 20% of students.
- **National University Entrance Exam (2018)** — ranked in the **top 1%** among 145,000+ participants.
- **National High School Entrance Exam (2013)** — admitted to NODET exceptional-talents high schools; ranked among top candidates.

TEACHING EXPERIENCE

Large Language Models (Dr. M. J. Dousti & Dr. Y. Yaghoobzadeh)	Spring 2026
Special Topics in NLP and LLMs (Dr. M. T. Pilehvar & Dr. Y. Yaghoobzadeh)	Fall 2025
Mathematics of Data Science (Dr. Mohammad Mahmoody)	Spring 2025
Large Language Models (Dr. M. J. Dousti & Dr. Y. Yaghoobzadeh)	Spring 2025
Machine Learning (Dr. Behnam Bahrak)	Fall 2024
Computational Intelligence (Dr. Nasser Mozayeni)	Fall 2022
Security of Computer Systems (Dr. Abolfazl Diyanat)	Fall 2022
Data Transmission (Dr. Abolfazl Diyanat)	Fall 2022
Computer-Aided Digital System Design (Dr. Meisam Abdollahi)	Spring 2022
Database Design (Dr. Hossein Rahmani)	Spring 2022
Security of Computer Systems (Dr. Abolfazl Diyanat)	Spring 2022
Data Structures (Dr. Hossein Rahmani)	Fall 2021
Algorithm Design (Dr. Sauleh Eetemadi)	Spring 2021
Database Design (Dr. Hossein Rahmani)	Spring 2021
Data Structures (Dr. Sauleh Eetemadi)	Fall 2020
Advanced Programming (Dr. Sauleh Eetemadi)	Spring 2020
Advanced Programming (Dr. Marzieh Malekimajd)	Fall 2019
Fundamentals of Programming (Dr. Zeinab Movahedi)	Fall 2019

ACADEMIC SERVICE

Reviewer

- ARR Emergency Reviewer, 2025–2026
- WiML Workshop, NeurIPS 2026
- MELT Workshop, COLM 2025
- NeurIPS Position Paper Track, 2025
- Queer in AI Workshop, NAACL 2024
- ImageArg Shared Task, 2023 — reviewer and program-committee member

TECHNICAL SKILLS

- **Programming:** Python, C#, C++, MATLAB, Verilog, VHDL, Assembly, Arduino
- **ML / DL & NLP:** PyTorch, Hugging Face Transformers, TRL, TensorFlow, Keras, scikit-learn, OpenCV; NumPy, Pandas, Matplotlib, Seaborn
- **MLOps & Deployment:** Docker, MLflow, Airflow, FastAPI, CI/CD, Kubernetes (basic)
- **Databases & Backend:** Microsoft SQL Server, Django
- **Tools & Workflow:** Git, Linux, $\text{\LaTeX}$, Azure, Scrum, ClickUp, Xilinx ISE, Proteus

SELECTED PROJECTS

- **MLOps Pipeline Portfolio** (2026) — end-to-end MLOps workflows: Dockerized & DVC-tracked ETL, Airflow orchestration, MLflow experiment tracking (8 tracked model runs), FastAPI model serving with request validation, and Kubernetes deployment strategies (rolling updates, blue-green, autoscaling) on a k3s cluster.
- **Large Language Models Course** (2024) — assignments on PEFT, quantization, RAG, RLHF, and PPO.
- **Applied Data Science Course** (2023) — end-to-end data pipelines combining classic and modern approaches.
- **Machine Learning Course** (2023) — classic ML across diverse settings; final project on image classification studying optimizers, augmentation, and learning-rate schedulers.
- **Medical Image Captioning & VQA** (2023) — fine-tuning the BLIP model for medical image captioning and visual question answering.
- **Computer Vision Course** (2022) — projects incl. document scanning (CamScanner-style), OCR, and tile-fraction detection.
- **Crawling the MedPix Database** (2022) — collected and organised open-access medical images and metadata, categorised by imaging plane.
- **Computational Intelligence Course** (2022) — from-scratch implementations of Kohonen SOFM, MLP, and Hopfield networks.
- **Natural Language Processing Course** (2022) — NLP fundamentals (tokenization, stemming, POS tagging) and classical/deep classification; image-captioning final project.
- **IUST Visual Question Answering** (2021) — four VQA model implementations in PyTorch.

CERTIFICATIONS

- **Applied MLOps Bootcamp** — Quera (2026)
- **Fundamentals of Agents** — Hugging Face Agents Course (Feb 2025)
- **Python for Data Science, AI & Development** — IBM Skills Network / Coursera (Jul 2021)
- **Machine Learning** — Stanford University / Coursera (May 2021)
- **Data Structures & Algorithms Specialization** — UC San Diego / Coursera: Data Structures, Algorithms on Strings, Algorithms on Graphs, Algorithmic Toolbox (2020)

LANGUAGES

Persian — Native • English — Professional working proficiency